

QUESTION 1 [20 marks]

(i) For the circuit shown in Figure 1,

- (15 marks) Calculate the equivalent resistance R_{eq} as seen from terminals a-b.
- (5 marks) Find the current i through the network using the result of part (a).

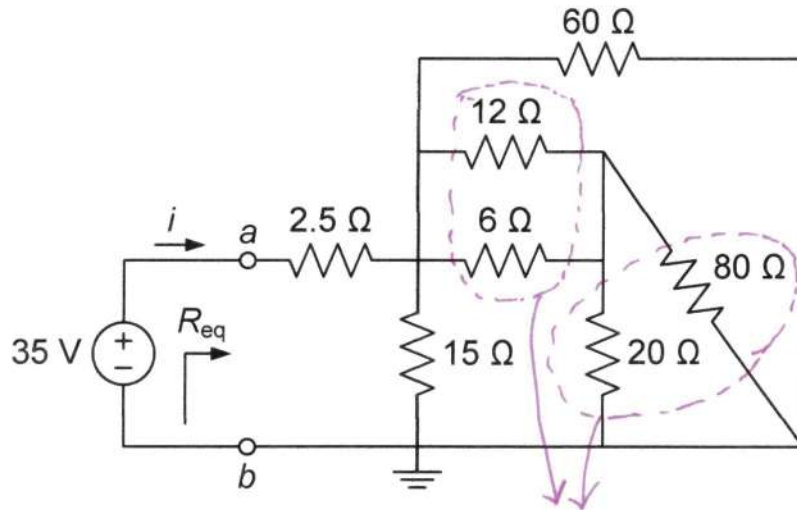
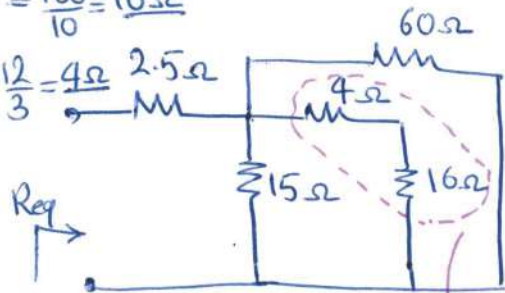


Figure 1

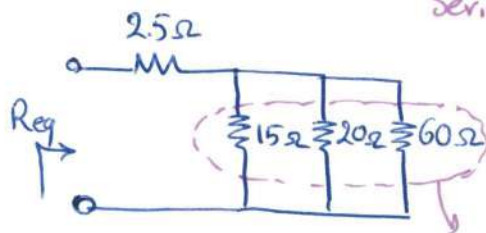
a)

$$20 \parallel 80 = \frac{20 \times 80}{100} = \frac{160}{10} = 16\Omega$$

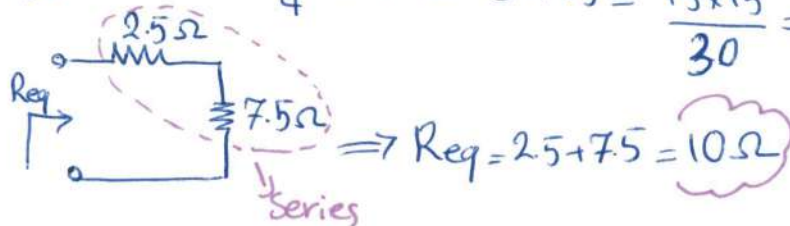
$$6 \parallel 12 = \frac{6 \times 12}{18} = \frac{12}{3} = 4\Omega$$



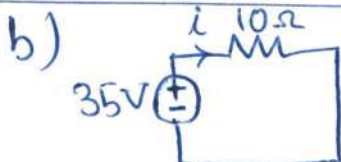
$$4 + 16 = 20\Omega$$



$$60 \parallel 20 \parallel 15 = \left(\frac{60 \times 20}{80} \right) \parallel 15 = \frac{60}{4} \parallel 15 = 15 \parallel 15 = \frac{15 \times 15}{30} = \frac{15}{2} = 7.5\Omega$$



$$\Rightarrow R_{eq} = 2.5 + 7.5 = 10\Omega$$



$$i = \frac{V}{R} = \frac{35}{10} = 3.5 \text{ A}$$

Ohm's Law

QUESTION 2 [40 marks]

(i) [24 marks] For the circuit shown in Figure 2,

- (10 marks) Apply nodal analysis to obtain the node voltage at nodes v_1 and v_2 , and show that $v_1 = 20\text{ V}$ and $v_2 = 12\text{ V}$
- (12 marks) Calculate all the powers absorbed/supplied by resistors and sources and specify which element supplies power and which element absorbs power.
- (2 marks) Verify the law of conservation of energy for this circuit.

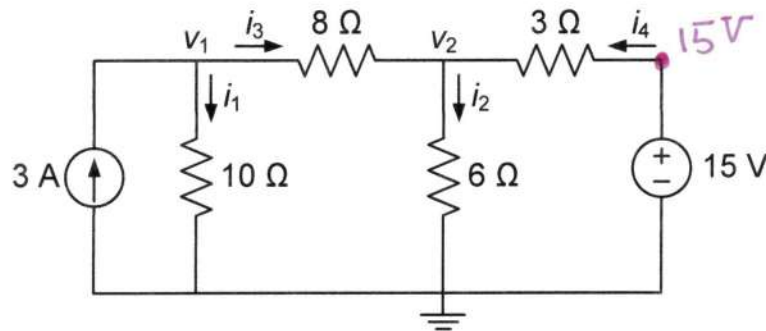


Figure 2

a) KCL @ v_1 : $3 = \frac{v_1}{10} + \frac{v_1 - v_2}{8}$ (based on directions of i_1 and i_3)

$\times 40 \rightarrow 120 = 4v_1 + 5v_1 - 5v_2 \rightarrow 9v_1 - 5v_2 = 120$

KCL @ v_2 : $\frac{v_1 - v_2}{8} + \frac{15 - v_2}{3} = \frac{v_2}{6}$ (based on directions of i_2, i_3, i_4)

$\times 24 \rightarrow 3v_1 - 3v_2 + 8 \times 15 - 8v_2 = 4v_2 \rightarrow 3v_1 - 15v_2 = -120 \times \frac{1}{3}$
 $\rightarrow v_1 - 5v_2 = -40$

$\Rightarrow \begin{cases} 9v_1 - 5v_2 = 120 \\ v_1 - 5v_2 = -40 \end{cases}$

$\textcircled{II} \rightarrow v_1 = 5v_2 - 40$ $\textcircled{II} \rightarrow 9(5v_2 - 40) - 5v_2 = 120 \rightarrow 45v_2 - 360 - 5v_2 = 120$
 $\rightarrow 40v_2 = 480 \rightarrow v_2 = \frac{480}{40} \rightarrow v_2 = 12\text{ V}$ $\textcircled{II} \rightarrow v_1 = 5 \times 12 - 40 = 60 - 40 = 20\text{ V}$
 $\rightarrow v_1 = 20\text{ V}$

b) $P = \frac{V^2}{R}$

- $10\Omega \rightarrow P_{10} = \frac{v_1^2}{10} = \frac{20^2}{10} = 40\text{ W absorbed}$
- $8\Omega \rightarrow P_8 = \frac{(v_1 - v_2)^2}{8} = \frac{(20 - 12)^2}{8} = \frac{8^2}{8} = 8\text{ W absorbed}$
- $6\Omega \rightarrow P_6 = \frac{v_2^2}{6} = \frac{12^2}{6} = 24\text{ W absorbed}$
- $3\Omega \rightarrow P_3 = \frac{(15 - v_2)^2}{3} = \frac{(15 - 12)^2}{3} = \frac{3^2}{3} = 3\text{ W absorbed}$

$P = V \cdot i$

3-A source: $P_{3A} = v_1 \times 3 = 20 \times 3 = 60\text{ W Supplied}$ (or -60W)

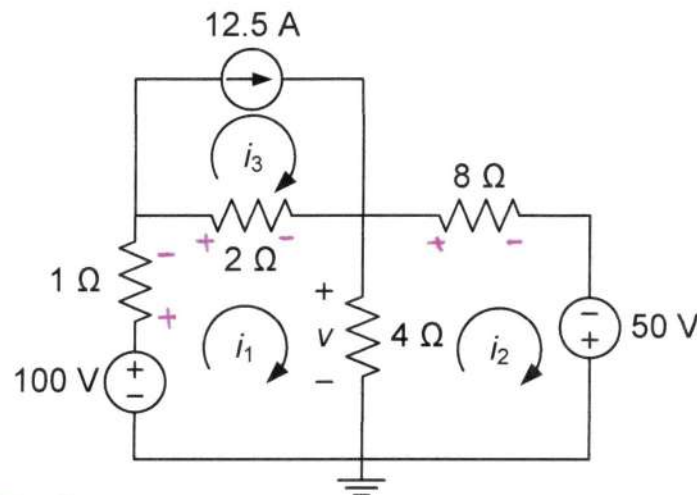
15-V source: $P_{15V} = 15 \times i_4 = 15 \times \frac{(15 - v_2)}{3} = 15 \times \frac{(15 - 12)}{3} = 15\text{ W Supplied}$ (or -15W)

c) Conservation of energy $\sum P = 0$
 $P_{10} + P_8 + P_6 + P_3 = P_{3A} + P_{15V}$
 $40 + 8 + 24 + 3 = 60 + 15$
 $75\text{ W} = 75\text{ W}$
 absorbed supplied

QUESTION 2

(ii) [16 marks] For the circuit shown in Figure 3,

- (10 marks) Apply mesh analysis to obtain the mesh currents i_1 , i_2 and i_3 , and show that $i_1 = 25$ A and $i_2 = 12.5$ A
- (2 marks) Find the voltage v across 4- Ω resistor.
- (4 marks) Calculate the power of the 50-V voltage source and explain whether it supplies or absorbs power.



a) mesh 3: $i_3 = 12.5$ A $\textcircled{I}$

Figure 3

KVL in mesh 1: $-100 + 1 \times i_1 + 2(i_1 - i_3) + 4(i_1 - i_2) = 0$
 $\textcircled{I} \rightarrow \underline{i_1 + 2i_1 - 2 \times 12.5 + 4i_1 - 4i_2 = 100} \rightarrow \underline{7i_1 - 4i_2 = 125}$

KVL in mesh 2: $-50 - 4(i_1 - i_2) + 8i_2 = 0 \rightarrow -4i_1 + 4i_2 + 8i_2 = 50$
 $\rightarrow -4i_1 + 12i_2 = 50 \xrightarrow{\times -\frac{1}{2}} \underline{2i_1 - 6i_2 = -25}$

$\textcircled{II} \Rightarrow \begin{cases} \textcircled{I} & 7i_1 - 4i_2 = 125 \\ \textcircled{II} & 2i_1 - 6i_2 = -25 \end{cases}$
 $\textcircled{III} \rightarrow i_1 = \frac{1}{2}(6i_2 - 25) = 3i_2 - 12.5 \xrightarrow{\textcircled{I}} 7(3i_2 - 12.5) - 4i_2 = 125 \rightarrow 21i_2 - 87.5 - 4i_2 = 125$
 $\rightarrow 17i_2 = 212.5 \rightarrow i_2 = \frac{212.5}{17} \rightarrow \underline{i_2 = 12.5 \text{ A}} \xrightarrow{\textcircled{II}} i_1 = \frac{1}{2}(6 \times 12.5 - 25) = \frac{1}{2}(75 - 25)$
 $i_1 = \frac{50}{2} \rightarrow \underline{i_1 = 25 \text{ A}}$

b) $v = 4(i_1 - i_2) = 4(25 - 12.5) = 4 \times 12.5 \Rightarrow \underline{v = 50 \text{ V}}$

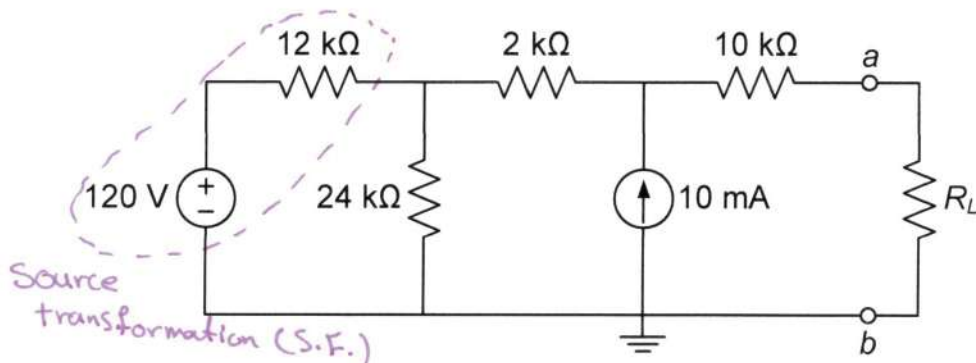
c) $P = v \cdot i = 50 \times i_2 = 50 \times 12.5 = 625 \text{ W supplied}$

(or -625W)

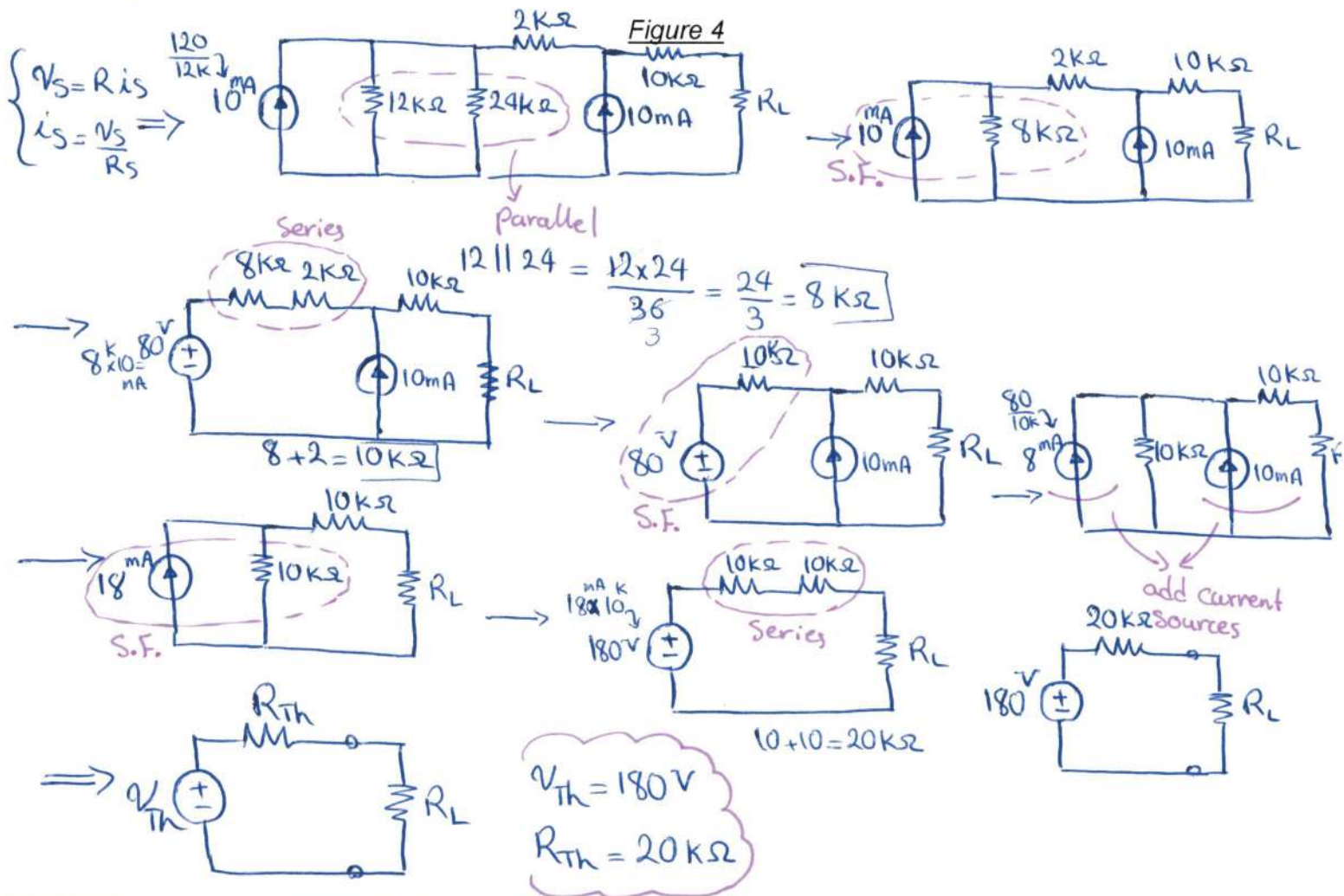
QUESTION 3 [20 marks]

(i) [10 marks] For the circuit shown in Figure 4,

- [6 marks] Use source transformation to obtain Thevenin equivalent circuit from terminals $a-b$ and draw the Thevenin equivalent circuit.
- [4 marks] Determine the value of load resistance R_L for maximum power transfer, and then calculate the maximum power that can be delivered to R_L .



a)

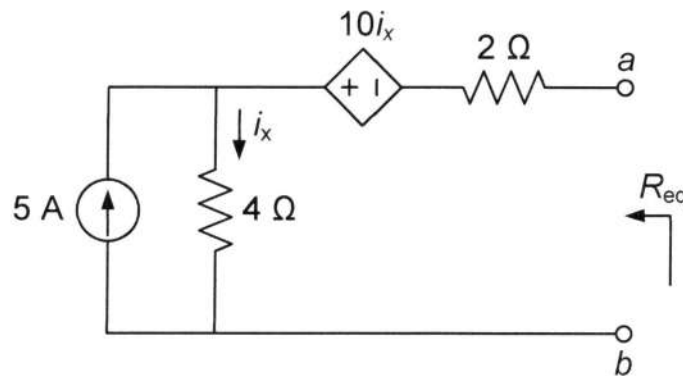


b) For max power transfer $\rightarrow R_L = R_{Th} \Rightarrow R_L = 20k\Omega$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(180)^2}{4 \times 20 \times 10^3} = 405 \text{ mW}$$

QUESTION 3

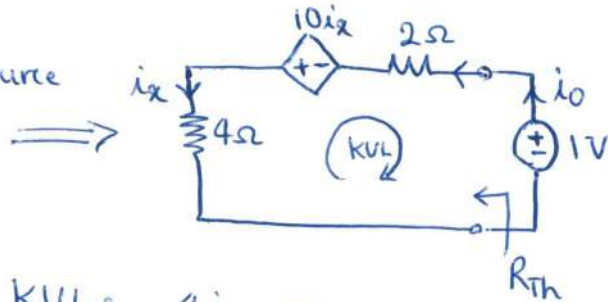
(ii) [10 marks] Find the equivalent resistance R_{eq} in the circuit given in Figure 5.



Dependent source $\rightarrow$ Connect an external Figure 5
voltage or current source

First approach: 1-V

* Connect a Voltage Source
 $V_o = 1V$
disconnect 5-A source



$$R_{Th} = \frac{V_o}{i_o} = \frac{1}{i_o}$$

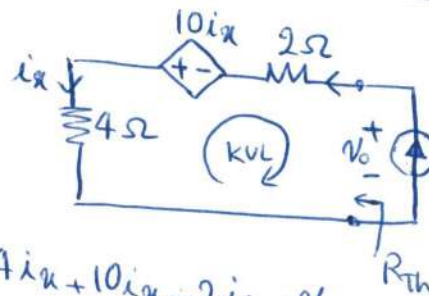
$$i_x = i_o \Rightarrow \text{KVL: } -4i_x + 10i_x - 2i_x + 1 = 0 \rightarrow 4i_x + 1 = 0$$

$$\rightarrow 4i_x = -1 \rightarrow i_x = -\frac{1}{4}A$$

$$\Rightarrow R_{Th} = \frac{1}{-\frac{1}{4}} = -4\Omega$$

Second approach:

* Connect a 1-A Current Source
 $i_o = 1A$
disconnect 5-A source



$$R_{Th} = \frac{V_o}{i_o} = \frac{V_o}{1}$$

$$i_x = 1A \Rightarrow \text{KVL: } -4i_x + 10i_x - 2i_x + V_o = 0 \rightarrow 4i_x + V_o = 0$$

$$i_x = 1A \rightarrow 4 + V_o = 0 \rightarrow V_o = -4V$$

$$R_{Th} = \frac{-4}{1} = -4\Omega$$

QUESTION 4 [20 marks]

- (i) In the circuit shown in Figure 6, the switch has been closed for a long time before it is opened at $t = 0$. The voltage source is given a step function.
- (4 marks) Find the initial voltage $v(0^-)$ across the capacitor under steady-state condition.
 - (2 marks) Calculate the initial energy $w_C(0)$ stored in the capacitor.
 - (4 marks) Find the final voltage $v(\infty)$ across the capacitor under steady-state condition.
 - (4 marks) Derive an expression for the voltage of the capacitor $v(t)$ for all time (i.e., for both $t < 0$ and $t > 0$).
 - (2 marks) Sketch the obtained voltage $v(t)$ in part (d) as a function of time.
 - (4 marks) Derive an expression for the current $i(t)$ through the $5\text{-}\Omega$ resistor for all time (i.e., for both $t < 0$ and $t > 0$).

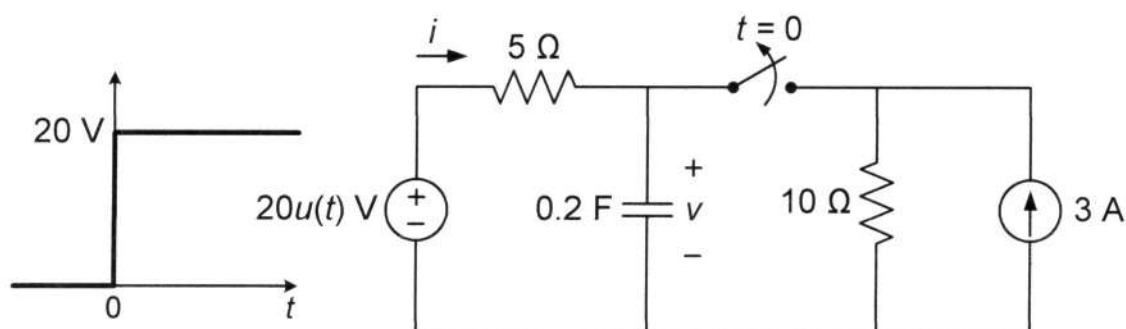
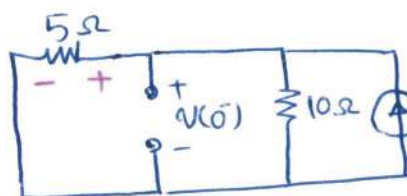


Figure 6

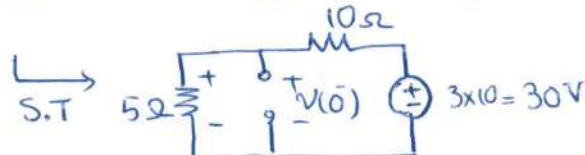
a) $20u(t) = \begin{cases} 0 & t < 0 \\ 20 & t > 0 \end{cases}$
step input voltage

$t < 0$ circuit before the change



$v(0^-) = V_{5\Omega} \text{ (parallel)}$

Circuit is in steady-state capacitor $\rightarrow$ open circuit

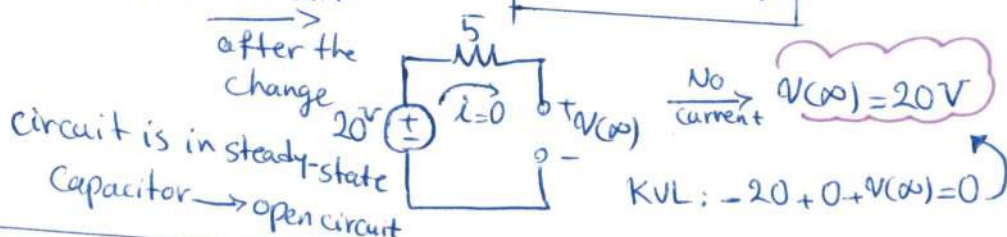


Voltage division

$v(0^-) = \frac{5}{5+10} \times 30 = 10\text{V}$

b) $w_C(0) = \frac{1}{2} C v^2(0) \rightarrow v(0) = v(0^-) \Rightarrow w_C(0) = \frac{1}{2} \times 0.2 \times 10^2 = 10\text{J}$

c) $t > 0$ circuit after the change



No sudden change in capacitor voltage

$v(\infty) = 20\text{V}$

KVL: $-20 + 0 + v(\infty) = 0$

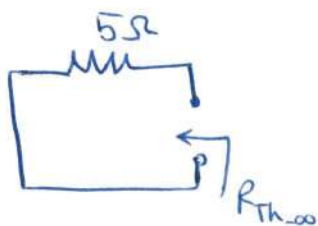
d) $v(t) = \begin{cases} v(0^-) & t < 0 \\ v(\infty) + (v(0^+) - v(\infty))e^{-t/\tau} & t > 0 \end{cases}$
for all time

$v(0^+) = v(0) = v(0^-) = 10\text{V}$

no sudden change after change

$$\tau = R_{Th\infty} \cdot C$$

Q4
Continued

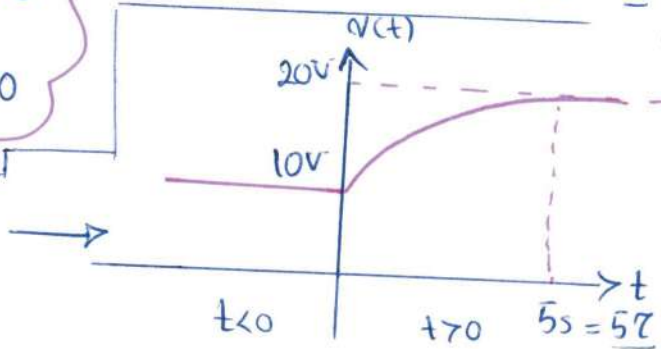


$R_{Th\infty}$: Thevenin equivalent resistance in the circuit from capacitor terminals after changes.

$$R_{Th\infty} = 5\Omega \rightarrow \tau = 5 \times 0.2 = 1s$$

$$\Rightarrow v(t) = \begin{cases} 10V & t < 0 \\ 20 - 10e^{-t} V & t > 0 \end{cases}$$

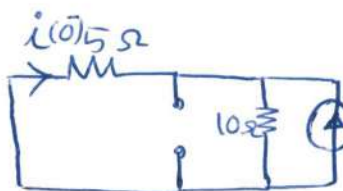
$$v(t) = 20 + (10 - 20)e^{-\frac{t}{\tau}} = 20 - 10e^{-\frac{t}{\tau}} \quad t > 0$$



e)

f) $t < 0$

Circuit before the change (steady-state)

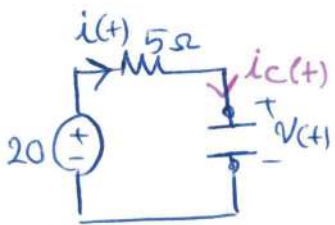


$$i(0) = -\frac{10}{5+10} \times 3 = -\frac{10}{5} = -2A$$

Current division

$t > 0$

Circuit after the change



* First approach:

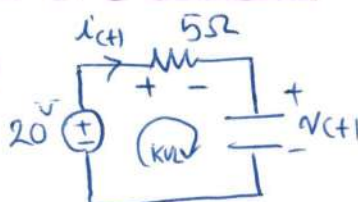
$$i(t) = i_C(t) = C \frac{dv(t)}{dt} = 0.2 \times (-10)(-1)e^{-t}$$

$$i(t) = \begin{cases} -2A & t < 0 \\ 2e^{-t} A & t > 0 \end{cases}$$

for all time

$$\rightarrow i(t) = 2e^{-t} A$$

* Second approach:



$$KVL: -20 + 5i(t) + v(t) = 0 \quad t > 0$$

$$\Rightarrow i(t) = \frac{20 - v(t)}{5} \quad t > 0$$

$$\Rightarrow i(t) = \frac{20 - 20 + 10e^{-t}}{5} = 2e^{-t} A \quad t > 0$$

$$\Rightarrow i(t) = \begin{cases} -2A & t < 0 \\ 2e^{-t} A & t > 0 \end{cases}$$